How Restrictive is U.S. Trade Policy?

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ABSTRACT

This short note computes Trade Restrictiveness Indexes for current U.S. trade policy. Building on the ideas of Anderson and Neary (1996, 2005), the Trade Restrictiveness Index is the uniform tariff that leaves the U.S. consumer as well off as under actual policy. As of September 2025, U.S. trade policy is twice as restrictive as headline tariff numbers suggest. The trade restrictiveness index is 23 percent, which stands in contrast to the 11 percent average tariff rate. Trade policy towards Canada and Mexico is two to three times more restrictive than average tariff rates suggest. Sectoral analysis shows that the restrictiveness is concentrated in vehicles, machinery, and electrical equipment.

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Introduction

Since January 2025, trade policy in the U.S. has changed dramatically with large and heterogeneous increases in tariff rates. This short note asks the following question: How restrictive is current U.S. trade policy? My answer: twice as restrictive as the headline numbers suggest.

Why the large gap? Two features of current policy make simple averages misleading. First, the heterogeneity in tariff rates—both across countries and products—means that simple average tariff rates will understate the overall restrictiveness of U.S. trade policy. The reason is that the deadweight loss from a tariff increases with the *square* of the tariff rate, not linearly. This implies that, in aggregate, two tariff regimes with the same average tariff but differences in the variance of tariffs have different economic consequences. Put simply: a few very high tariffs create more distortions than many moderate tariffs with the same average rate.

To address this challenge, I utilize the ideas of Anderson and Neary (1996, 2005), who index how restrictive trade policy is by the uniform tariff that, when applied to all goods, leaves the U.S. consumer as well off as under the actual, heterogeneous tariffs. In general, implementing this approach is not a trivial task and requires a complete specification of the data generating process. Instead, I employ the quadratic approximation developed by Feenstra (1995) and Kee, Nicita, and Olarreaga (2009), resulting in an easily implementable measure of trade restrictiveness which is the trade-weighted, root-mean-square of tariffs. Irwin (2010) applies this framework to historical U.S. tariffs; I extend the approach to current policy.

The second challenge is the following—what *is* U.S. trade policy? The issue is that a simple, numerical statement about the tariff is sometimes not possible. For example, whether a U.S. importer from Mexico faces a tariff depends upon whether that importer has shown that the product is covered under the United States-Mexico-Canada Trade Agreement (USMCA). As another example, many products imported from Brazil are tariffed or not depending upon whether they ultimately go into civilian aircraft. The steel and aluminum tariffs extend to derivative products, but only their steel and aluminum content is tariffable. And the list goes on.

To address this concern, I take a "revealed preference" approach by inferring what tariffs firms actually pay from the data, rather than what the law says they should pay. Specifically, I use the public Census trade data, which reports duty collected and values at the 10-digit Harmonized System level, by country. This approach allows me to infer the *de facto* applied tariff rate for each product-country pair. This is in contrast to trying to measure the *de jure* tariffs by following the announced statutes and then making assumptions about how they might be implemented.[1]

Using the de facto tariffs, I compute trade restrictiveness indices in aggregate and by U.S. trading partner. Two main results emerge from this analysis. First, in aggregate, current U.S. trade

[1] Waugh (2025) (`https://www.tradewartracker.com`) or The Budget Lab at Yale (2025) are examples that try and construct estimates of the weighted average tariff given the announced statutes.

policy (as of September 2025) has a trade restrictiveness index of 23 percent. That is, a 23 percent uniform tariff yields the same deadweight loss as current trade policy. In contrast, the average tariff rate (tariff revenue divided by imports) is 11 percent. Current U.S. trade policy is more restrictive — by thirteen percentage points — than it appears.

The second important result is that for certain countries, U.S. trade policy is significantly more restrictive than what it appears to be. Canada and Mexico are the most important examples, with trade restrictiveness indices of 12 and 10 percent, respectively. In contrast, average tariff rates are 4 and 5 percent.

The issue here is that most trade with these countries faces zero tariffs under the USMCA, but the tariffs that are applied — concentrated on products like autos and metals — are important and create substantial distortions that simple averages fail to capture. I confirm this point by computing Sectoral TRIs and find that vehicles face a high level of trade restrictiveness equivalent to a uniform tariff of roughly 23 percent.

These distinctions matter for policy. Trade restrictiveness indices are used by international institutions to monitor trade policy and by negotiators to evaluate whether proposed concessions represent genuine liberalization or not. A country can lower average tariffs while maintaining a high TRI by offering reductions on low-volume products while keeping high tariffs on key sectors. Current U.S. trade policy towards Canada and Mexico is a case in point. And the sectoral TRIs make clear that high tariffs are concentrated in specific sectors.

1. Trade Restrictiveness in Theory

This section describes the theoretical framework behind the measurement of the trade restrictiveness indices presented in Section 2. The framework below builds on ideas in Anderson and Neary (1996, 2005), who define a Trade Restrictiveness Index (TRI) as the uniform tariff that, when applied to all goods, leaves the representative household as well off as under the actual vector of trade distortions.

As discussed in Kee et al. (2009), the Anderson–Neary measure faces several difficulties in practice. In line with their approach, I construct a second-order, welfare-consistent approximation to the TRI, relying on a Harberger-style quadratic expansion of the welfare loss. This derivation is closely related to the treatment in Feenstra (1995). This approximation avoids the computational burden of the full Anderson–Neary CGE approach, which requires specifying production functions, preference parameters, factor markets, and numerically solving for general equilibrium prices. Instead, the approximation requires only observable data on tariffs, import shares, and (possibly) elasticities of import demand.

1.1. The Deadweight Loss of Tariffs

The environment considered is one in which there are many goods indexed $g = 1, \dots, G$. The government levies ad valorem tariffs $\tau_{g,t}$ on each good, at date t. The framework applies to any definition of a "good." In the empirical implementation, I define goods at the HS10-by-country level, treating the same product from different countries as distinct goods facing potentially different tariffs.

The goal is to summarize the entire vector $\{\tau_{g,t}\}_{g=1}^{G}$ with a *single* uniform tariff rate $\hat{\tau}_t$, such that the uniform tariff has the same welfare impact as the structure of actual tariffs. To operationalize this idea, I employ a Harberger-style deadweight-loss formula, first for an individual good and then aggregated across all goods.

The Harberger triangle formula associated with a tariff on good g is

$$\Delta W_{g,t} \approx \frac{1}{2}\,\theta_{g,o}\,\tau_{g,t}^2\,M_{g,o}, \tag{1}$$

where $M_{g,o}$ is import expenditure and $\theta_{g,o}$ is the (absolute value of the) compensated import demand elasticity, all evaluated at the initial equilibrium with the subscript o indexing initial values. And to reiterate, the subscript t is indexing the policy at date t. In the Appendix, I show that an alternative derivation based on a second-order expansion of the expenditure function yields the same results.

I then aggregate (1) across all goods and arrive at the **aggregate** deadweight loss from the tariff vector $\{\tau_{g,t}\}$

$$\Delta W_t \approx \frac{1}{2}\sum_{g=1}^{G}\theta_{g,o}\tau_{g,t}^2 M_{g,o}. \tag{2}$$

This deadweight loss calculation can then be expressed as a weighted average of the square of tariffs:

$$\Delta W_t \approx \frac{1}{2}A_o\sum_{g=1}^{G}\omega_{g,o}\tau_{g,t}^2, \tag{3}$$

$$\text{where} \quad A_o \equiv \sum_{g'=1}^{G}\theta_{g',o}M_{g',o} \quad \text{and} \quad \omega_g \equiv \frac{\theta_{g,o}M_{g,o}}{\sum_{g'=1}^{G}\theta_{g',o}M_{g',o}}.$$

The weights in (3) reflect two issues: (i) how sensitive good g is to a change in price — this is the elasticity $\theta_{g,o}$ part; and (ii) how important good g is as measured by import expenditure $M_{g,o}$.

1.2. The Uniform Tariff Equivalent

Following the Anderson–Neary logic, I want to find the uniform tariff — $\hat{\tau}_t$ — which leads to the same aggregate deadweight loss as in (3). Under a uniform tariff, the deadweight loss is

$$\Delta W_t^{uni} \approx \frac{1}{2} A_o \, \hat{\tau}_t^2. \tag{4}$$

where the A_o term is defined the same as in (3). Then equating (4) and (3) gives

$$\hat{\tau}_t = \sqrt{\sum_{g=1}^{G} \omega_{g,o} \tau_{g,t}^2}, \tag{5}$$

where the uniform tariff equivalent is the weighted, root-mean-square tariff.

I'll make one more assumption to simplify things — I'll abstract from heterogeneity in elasticities, so $\theta_{g,o} = \bar{\theta}$. This is a strong assumption and deviates from Kee et al.'s (2009) approach, which estimates good-specific elasticities. The appeal here is that measurement is simplified dramatically. Under the common import demand elasticity assumption, the uniform tariff simplifies to

$$\hat{\tau}_t = \sqrt{\sum_{g=1}^{G} s_{g,o} \tau_{g,t}^2}, \qquad \text{where} \quad s_{g,o} \equiv \frac{M_{g,o}}{\sum\limits_{g'=1}^{G} M_{g',o}}. \tag{6}$$

Under this assumption, the uniform tariff is an import-share-weighted, root-mean-square of actual tariffs. This provides a simple, back-of-the-envelope index that captures in one number how restrictive or not a tariff regime is and in a theoretically consistent way. For the rest of the paper, I'll refer to $\hat{\tau}$ in (6) as the Trade Restrictiveness Index (TRI).

1.3. Alternative Aggregate Tariff Measures

I'll contrast the TRI with two alternative tariff measures. The first one is the mean weighted tariff which is

$$\bar{\tau}_t = \sum_{g=1}^{G} s_{g,o} \tau_{g,t} \tag{7}$$

where the weights s_g are the same as in the TRI measure in (6). Under the assumption of common import demand elasticities, the mean weighted average tariff can also be interpreted as Anderson and Neary's (2003) Mercantilist Index of Trade Policy (MTRI). The MTRI answers the question: what uniform tariff would leave aggregate imports at their current level? As shown in Kee et al. (2009), the MTRI is just a weighted average tariffs where the weights are the same as discussed in (3). And with common import demand elasticities, the weights are simply the import shares.

The TRI connects with the mean weighted tariff through the following relationship

$$\hat{\tau}_t = \bar{\tau}_t \sqrt{1 + \sigma^2_{\tau,t}}, \quad \text{where} \quad \sigma^2_{\tau,t} = \sum_{g=1}^{G} s_{g,o} \left(\frac{\tau_{g,t} - \bar{\tau}_t}{\bar{\tau}_t} \right)^2 \tag{8}$$

where the $\sigma_{\tau,t}$ term is the import-weighted variance of tariffs, measured as percent deviations from the mean. This means that for a given average tariff, a regime with high tariff dispersion — such as one with many exemptions and a few very high rates — will be substantially more restrictive than the average suggests. As I discuss in Section 3, this is precisely the situation with current U.S. policy toward Canada and Mexico.

Another commonly used metric is the total value of duties collected divided by the total value of imports or

$$\tilde{\tau} = \frac{\sum_{g=1}^{G} M_{g,t}\tau_g}{\sum_{g'=1}^{G} M_{g',t}} \quad = \quad \sum_{g=1}^{G} s_{g,t}\tau_{g,t}. \tag{9}$$

The last line makes clear that this is a weighted average tariff — just that the weights are varying with the date (the s's are indexed by t, not the original pre-tariff point at o). While widely used, it is problematic because the weights will typically fall on the goods with the largest tariffs as firms and consumers substitute away. Thus, this measure will tend to understate how restrictive trade policy is relative to (7) and (6).

2. Measuring Trade Restrictiveness

This section details how I measure the objects in (6) in the data. My main data source is the U.S. Census FT900 U.S. International Trade in Goods and Services data release and specifically the Monthly International Trade Datasets. These data provide detail on U.S. imports and the duty collected at the Harmonized System (HS) 10 digit level, by country, at the monthly frequency. The Census API only has the Monthly datasets dating back to 2013 and thus is the start of the time series that I consider.

2.1. Measuring Tariffs and Weights

As mentioned in the introduction, my main approach is to infer tariffs from the data rather than follow the statutes and make assumptions about how they are implemented or the extent to which some firms are subject to tariffs or not. Specifically, I measure tariffs by computing

$$\tau_{g,t} = \frac{\text{duties collected on imports } g \text{ for consumption}_t}{\text{value of imports of } g \text{ for consumption}_t}, \tag{10}$$

or the ratio of the dutiable value of imports for consumption relative to the value of imports for consumption. The inferred tariff represents the average effective tariff rate actually paid by importers of good g. This corresponds to $\tau_{g,t}$ in the theoretical framework, where the relevant tariff for welfare calculations is the rate importers actually face, not the statutory rate that may or may not apply depending on exemptions.

The definition of a good g will depend upon the context, so when I analyze overall trade restrictiveness it will be a country-by-HS10-code. When I analyze country-specific TRIs, the country dimension is no longer relevant and it is just an HS10-code.

What this approach will do is infer the extent to which some importers are able to claim exemptions, as well as the tariffable content embedded in the "typical product." For example, importers that are non-USMCA compliant would face a tariff of 25 percent when importing from Mexico. In this example, when I infer the tariff, it will reflect an average of (i) importers who are compliant and thus face the USMCA tariff and (ii) importers who are not-compliant and face the 25 percent tariff. The data reveals the division between (i) and (ii), not myself or other researchers making this ex-ante choice.

The weights $s_{g,o}$ are measured in a similar way. They are the annual value of imports for a good divided by the total annual value of all imports in the year 2024. As mentioned above, a good is either the cross of a country by HS10 code or just an HS10 code in the country-specific contexts.

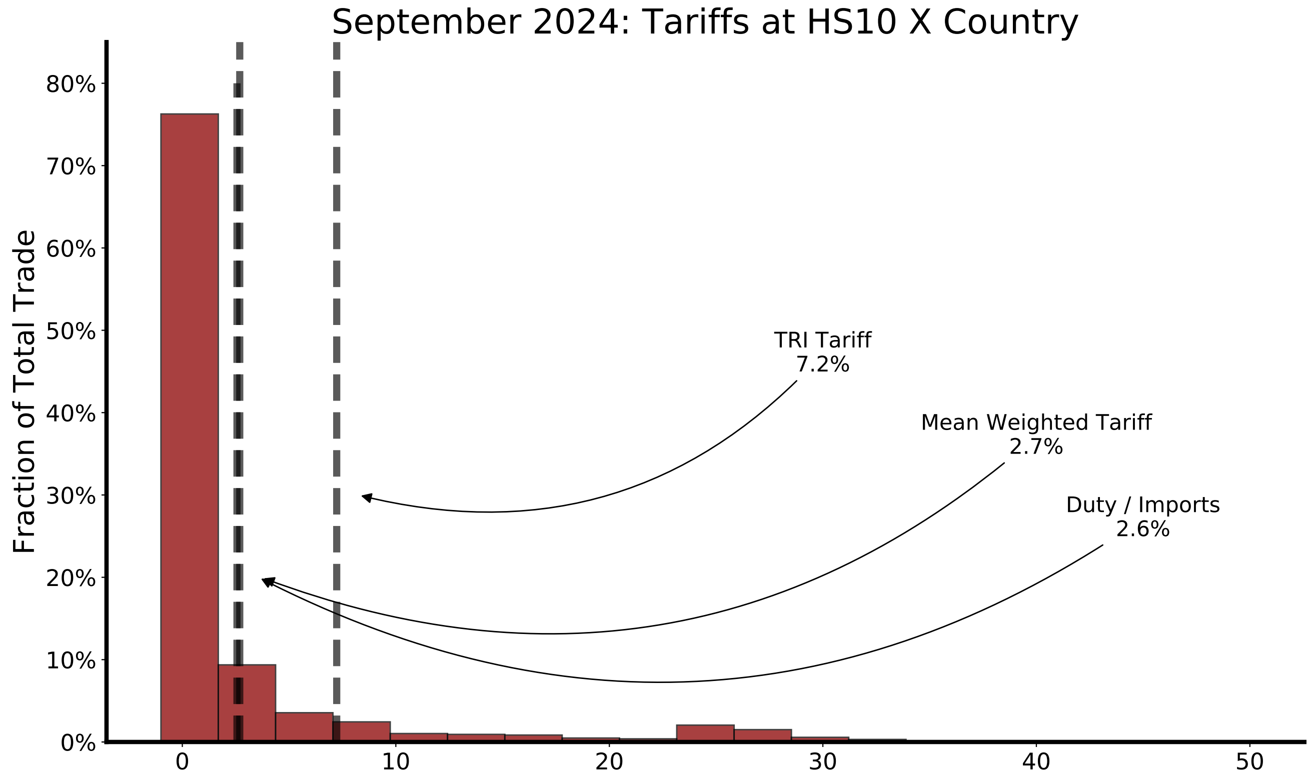
3. The Restrictiveness of U.S. Trade Policy

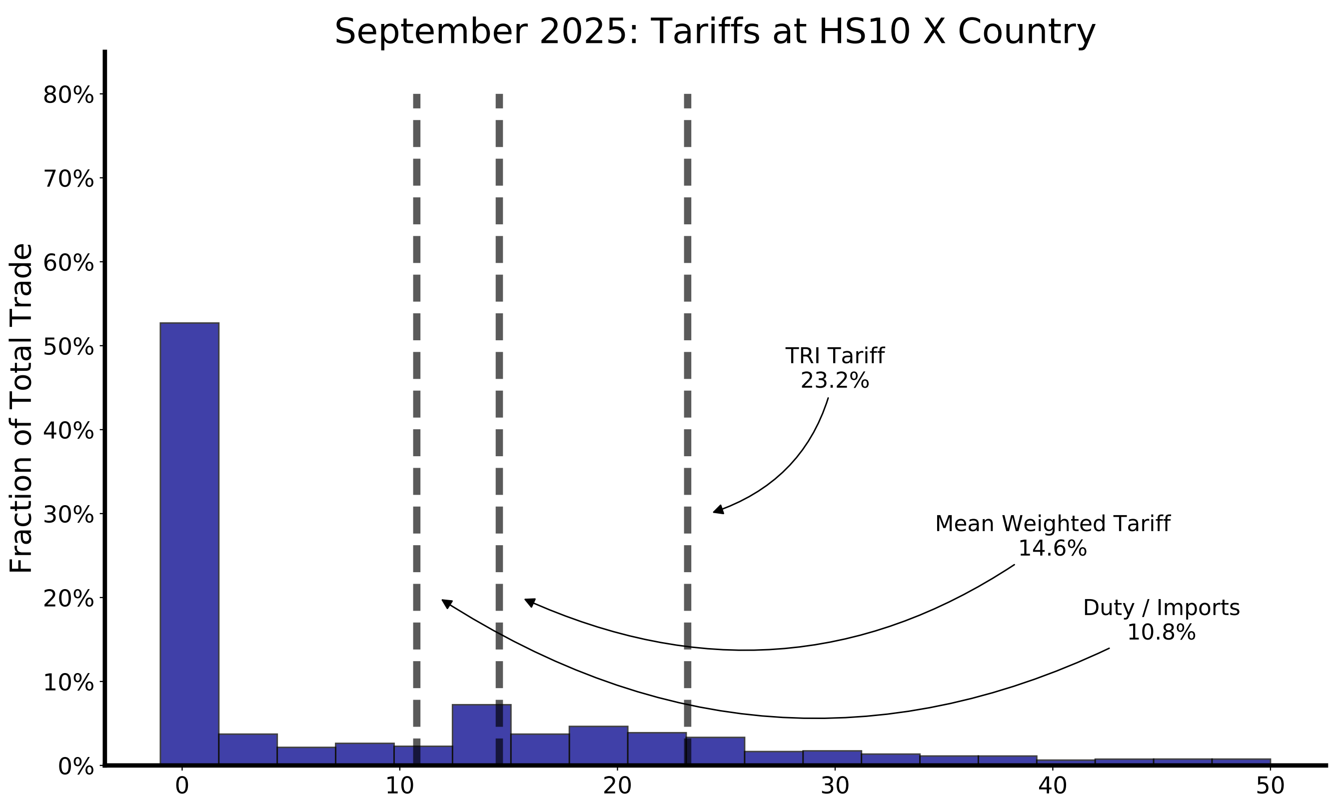
Figure 1 shows how restrictive U.S. trade policy has become. In both panels, I report a histogram of tariffs for two time periods: September 2024 and September 2025. The unit of observation here is the tariff associated with an HS10 code and country. The y-axis reports the share of U.S. imports that lies within each particular bin. Three vertical lines are shown which demarcate the three aggregate tariff measures discussed above: (i) the TRI tariff defined in (6), (ii) the mean weighted tariff in (7), and (iii) duty divided by imports in (9).

In September 2024, mean tariffs were low at 3 percent and nearly 75 percent of trade was being imported with near zero tariffs. However, there is meaningful dispersion in tariffs for the remaining 25 percent of imports. And this dispersion leads to a TRI tariff measure of 7 percent.

The bottom panel of Figure 1 reports the most recent month available, September 2025. It starkly illustrates how much more restrictive U.S. trade policy has become.

Figure 1: The Distribution of Tariffs (HS10 by Country)





The most important observation is that U.S. trade policy is substantially more restrictive than the mean tariff rate would suggest. The TRI tariff currently stands at 23 percent, more than 12 percentage points larger than what the simple duty / imports tariff rate of 11 percent would imply, and 9 percentage points larger than the mean weighted tariff of 15.[2] Recall that the TRI is the uniform tariff that would generate the same aggregate welfare loss as the actual tariff structure. A TRI of 23 percent means that current policy — with its mix of exemptions and high rates — is as costly to U.S. consumers as a uniform 23 percent tariff on all imports.

The key issue is that current U.S. trade policy is generating a large amount of dispersion in tariff rates, even though the average tariff rate seems modest. The histogram illustrates the point by highlighting how large amounts of U.S. imports are subject to tariff rates exceeding 10 or even 20 percent. The TRI tariff recognizes that the welfare cost of these higher tariff rates increases quadratically. Thus, the uniform tariff required to deliver the same deadweight loss as the current tariff regime is significantly larger than what other tariff measures would imply.

3.1. Over time and Across Countries

The histograms in Figure 1 hide interesting variation both across time and countries. Figure 2 illustrates this point by plotting different tariff measures for the top 3 U.S. trading partners since the beginning of 2025 and the All Country aggregate. Table 1 reports tariff metrics for the top 20 U.S. trading partners as of September 2025.

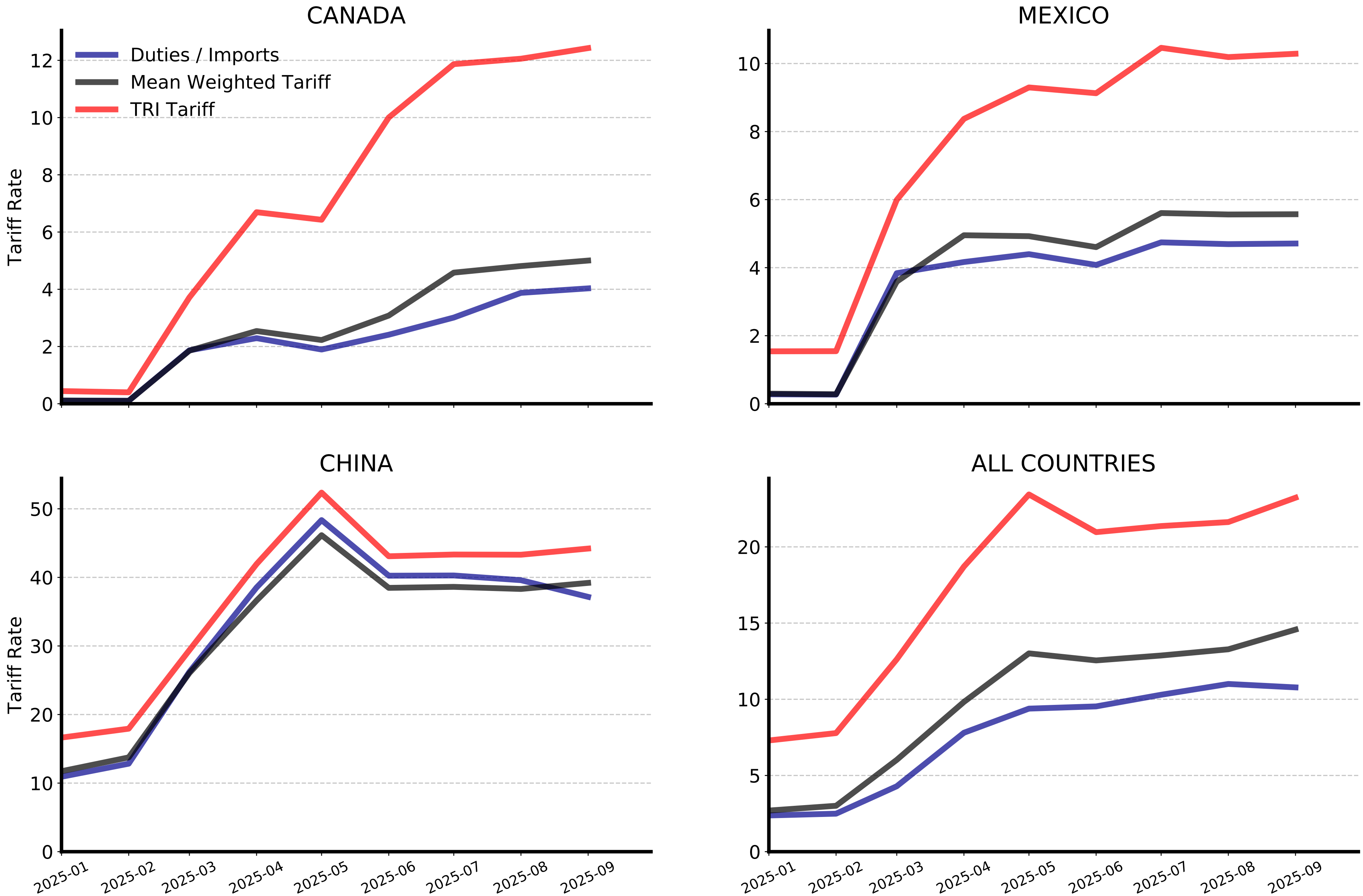
The top-left panel of Figure 2 shows that Canada is an important case where the TRI has increasingly diverged from average measures of tariff rates. Since March 2025, average tariff rates were near zero and have risen to 4 percent by September 2025. In contrast, the TRI tariff measure is 12 percent — U.S. policy towards Canada is three times more restrictive than what average tariff rates suggest.

The divergence between the TRI and average tariffs for Canada and Mexico reflects the structure of tariff policy since March 2025. USMCA-compliant goods were exempted from the 25 percent IEEPA tariffs, allowing most trade to continue duty-free. However, automobiles face a 25 percent tariff under Section 232 regardless of USMCA status, and steel and aluminum face tariffs of 50 percent. This combination — lots of trade at zero, but key categories facing high tariffs — generates exactly the high-dispersion regime that the TRI is designed to capture.

The same issue is apparent for Mexico. Like Canada, average tariff rates have been relatively low. But the large TRI tariff means that U.S. trade policy is much more restrictive towards Mexico — in this case nearly two times more restrictive. Also like Canada, visual inspection

[2] As a separate observation, the mean weighted tariff does not differ markedly from measures using the announced statutory rates. For example, Waugh or The Budget Lab at Yale report weighted tariff rates of between 15 and 17 percent depending upon the exact month looked at versus 15 using tariffs inferred from the Census data.

Figure 2: Tariff Measures for Selected Countries



of the data shows that autos and auto parts are important categories facing large tariffs, while many imports are coming in near duty free.

China is the opposite, with a relatively small and stable difference between the TRI tariff rate and average tariffs. The distinction here is that there is simply less variance in tariff rates across goods relative to the Canadian (or Mexican case). Yes, the overall level of tariffs is much higher. But tariffs vis-a-vi China don't face the same scale of exemptions that imports from Canada or Mexico do, and thus the TRI tariff measure differs little relative to average tariffs.

Table 1 provides an overview of these tariff metrics as of September 2025 for the top 20 U.S. import sources plus the All Country aggregate. In addition to the examples discussed above, Brazil and Switzerland are other countries that stand out with large differences between trade restrictiveness and simple average tariffs.

3.2. Trade Restrictiveness By Sector

The TRIs in (6) can also be constructed at sector-level. The sectoral TRI answers the following question: What uniform tariff applied to all goods in a sector would generate the same deadweight loss as the actual tariff structure within that sector?

In constructing the sectoral TRI, a good is defined as a country by HS10 code. The sectoral definition that I focus on is two-digit HS codes. In Figure 3, I focus on four sectors: Machinery (HS code 84), Electrical Equipment (HS code 85), Vehicles (HS code 87), and Pharmaceuticals (HS code 30). Excluding petroleum and precious metals, these four sectors account for 50 percent of all U.S. imports.

Figure 3 presents the results. As with the other TRIs, overall restrictiveness is large in three out of four categories. As of September 2025, trade restrictiveness in Machinery, Electrical Equipment, and Vehicles is more or less equivalent to a 23 percent tariff.

There is, however, heterogeneity in the magnitude relative to what average tariff measures would suggest. In Machinery, the simple measure of duties relative to imports would suggest trade is only modestly restricted with a tariff rate of 8 percent, substantially below the TRI of 24 percent. Electrical Equipment displays a similar pattern.

Vehicles is a counterexample. Here there is less of a gap between the TRI and either the mean weighted tariff of 19 or duty measure of 18. This reflects the structure of the Section 232 auto tariffs, which apply broadly across the sector regardless of USMCA status. Unlike Machinery or Electrical Equipment — where cross-country heterogeneity and various exemptions create a mix of low-tariff and high-tariff goods — most vehicle imports face similar tariff rates. Lower dispersion within the sector implies that the TRI and average measures are more similar.

The Vehicle results help explain the country-level patterns documented in Section 3.1. Canada and Mexico are major sources of U.S. vehicle imports, and the high sectoral TRI for Vehicles contributes directly to the elevated TRIs for these countries. At the same time, much of their non-auto trade enters duty-free under USMCA, which keeps average tariffs low. The combination of high sectoral restrictiveness in autos and near-zero tariffs elsewhere generates exactly the dispersion that drives the large TRI for Canada and Mexico.

The Pharmaceuticals category is where there is a massive difference between the TRI and average measures of tariffs. Overall restrictiveness is modest at 5 percent, but this is still roughly five times higher than the duty measure suggests. This result lies behind Ireland's low TRI, but large gap between the TRI and average measures (see Table 1). Approximately half of U.S. imports from Ireland in 2024 are in the Pharmaceutical category.

Another interesting observation is that the scope of substitution or changes in sourcing patterns differs between sectors. This can be seen in the divergence between the mean weighted tariff and the duty measure. As discussed above, the key distinction between these two measures is that for the mean weighted tariff, the weights are fixed, whereas for the duty measure the weights are changing. Thus, for Vehicles the similarity between the mean weighted and duty measure suggests that changes in sourcing patterns or substitution have been small. However, Machinery and Electrical Equipment display divergence between the mean weighted and duty measure suggesting that substitution is stronger in these categories.

As a summary, Table 2 reports the sectoral tariff measures along with informative names and the sector's share of U.S. imports in 2024, excluding petroleum and precious metals.

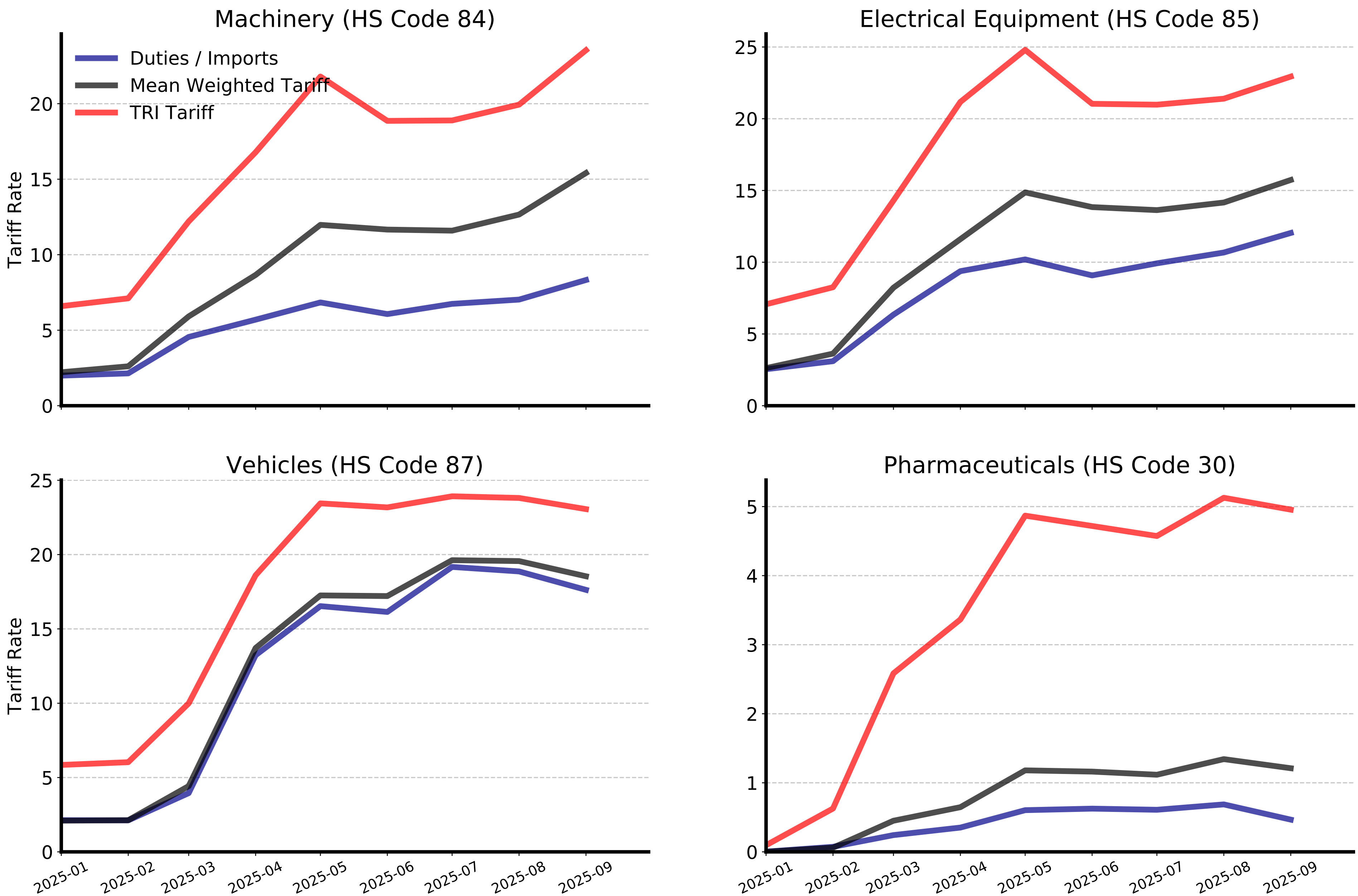
4. Conclusion

Since January 2025, it's become popular for economists, journalists, and policy makers to discuss and track aggregated tariff numbers regarding the current state of U.S. trade policy. However, different tariff regimes are not all created equal — even conditional on having the same average rate of protection.

This short note used the ideas of Anderson and Neary (1996, 2005) to construct a theoretically grounded measure as to how restrictive U.S. trade policy is. Specifically, I measured the uniform tariff that leaves the U.S. consumer as well off as under actual policy — both in aggregate, by trading partner, and by sector. As the results show, policy has become far more restrictive than what the headline numbers suggest. And these results imply that policy assessments based on average tariffs may be misleading.

With that said, there are important limitations of the analysis. While the measurement is theoretically grounded, I do employ approximations and abstract from details like goods-specific elasticities of trade which may be important. Trade restrictiveness indices only measure how

Figure 3: Tariff Measures for Selected HS2 Sectors



restrictive policy is; they do not quantify the potential inflationary and output costs.

Finally, this is a live document. As newer data are released, the paper here will update and present updated versions of the results. And the website `https://www.tradewartracker.com` will have interactive visualizations of the tariff measures presented here.

Table 1: Tariff Metrics by Country: September 2025

Country	TRI Tariff	Mean Weighted Tariff	Simple Mean Tariff
China	44.2	39.2	37.2
Brazil	32.3	19.9	17.3
India	26.6	18.5	17.5
All Countries	23.2	14.6	10.8
Indonesia	22.3	18.6	19.5
Vietnam	21.3	16.3	12.4
Korea, South	20.9	17.3	15.8
Japan	20.4	16.7	16.8
Thailand	19.4	15.0	9.7
Switzerland	18.8	9.4	6.7
Germany	17.7	13.2	12.6
Taiwan	16.3	9.4	5.1
Italy	15.6	11.2	8.5
Malaysia	13.9	9.2	10.1
France	13.3	9.2	8.8
Netherlands	12.9	7.0	8.7
Canada	12.4	5.0	4.0
Mexico	10.3	5.6	4.7
United Kingdom	8.7	5.9	5.6
Singapore	5.8	2.3	2.6
Ireland	4.9	1.6	0.4

Table 2: Tariff Metrics by Sector: September 2025

HS2	HS2 Name	TRI Tariff	Mean Weighted Tariff	Simple Mean Tariff	Share of U.S. Imports
73	Iron/Steel Articles	49.9	42.4	41.2	1.5
76	Aluminum	46.2	40.4	37.8	0.8
61	Knitted Apparel	45.0	42.1	39.4	1.4
62	Woven Apparel	40.9	36.7	31.6	1.1
72	Iron & Steel	38.1	31.0	30.4	1.0
64	Footwear	34.9	33.0	31.4	0.8
94	Furniture	34.1	24.3	19.2	2.1
95	Toys & Games	29.3	27.5	25.0	1.3
39	Plastics	25.8	17.2	15.5	2.2
40	Rubber	25.3	18.7	16.4	1.1
84	Machinery	23.5	15.4	8.3	15.8
87	Vehicles	23.1	18.5	17.6	12.0
85	Electrical Equipment	22.9	15.7	12.0	14.5
90	Optical/Medical Equipment	19.0	13.3	11.5	3.8
29	Organic Chemicals	14.8	6.5	1.4	1.7
38	Miscellaneous Chemicals	14.2	7.9	12.1	0.7
88	Aircraft	13.1	7.3	7.8	1.1
22	Beverages & Vinegar	11.1	6.9	6.3	0.9
08	Fruit & Nuts	7.5	1.8	3.8	0.7
30	Pharmaceuticals	5.0	1.2	0.5	6.7

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Appendix: Expenditure-Function Approach to Trade Restrictiveness

This appendix outlines the duality-based approach behind the Anderson-Neary Trade Restrictiveness Index (TRI) and shows how a second-order Taylor expansion of the expenditure function yields the same root-mean-square (RMS) type expression in (6).

A. Expenditure Function and Price Distortions

For clarity, I'll abstract from the time index in the presentation below. Now define $e(p, u)$ as the expenditure function: the minimum cost of attaining utility u at prices p. If the government imposes ad valorem tariffs $\tau = (\tau_1, \ldots, \tau_G)$, consumer prices are

$$p_g^{\tau} = p_g(1 + \tau_g). \tag{11}$$

Equivalent variation of the tariff vector is

$$EV(\tau) = e(p^{\tau}, u^o) - e(p, u^o), \tag{12}$$

where u^o is the pre-tariff utility level. Now the *uniform tariff* $\hat{\tau}$ is said to be *welfare-equivalent* if

$$e(p(1 + \hat{\tau}), u^0) = e(p^{\tau}, u^0), \tag{13}$$

which is the defining condition of the Trade Restrictiveness Index.

To obtain an analytically tractable expression, expand the expenditure function around the initial prices p:

$$e(p + \Delta p) \approx e(p) + \sum_g h_g(p)\, \Delta p_g + \frac{1}{2} \sum_{g,g'} S_{gg'}(p)\, \Delta p_g \Delta p_{g'}, \tag{14}$$

The term $h_g(p) = \partial e / \partial p_g$ is Hicksian (compensated) demand evaluated at initial prices p, which follows from Shephard's Lemma. And then the other term: $S_{gg'}(p) = \partial^2 e / \partial p_g \partial p_{g'}$ is an element in the Slutsky substitution matrix. And I'm suppressing the notational dependence on the initial level of utility u^o.

Now $\Delta p_g = p_g \tau_g$. Thus the welfare change associated with the full tariff vector is approximately

$$EV(\tau) \approx \sum_g h_g p_g \tau_g \; + \; \frac{1}{2} \sum_{g,g'} S_{gg'} p_g p_{g'} \tau_g \tau_g'. \tag{15}$$

Evaluating the same expansion for a uniform tariff $\hat{\tau}$ gives

$$EV^{uni}(\hat{\tau}) \approx \hat{\tau} \sum_g h_g p_g \; + \; \frac{1}{2}\hat{\tau}^2 \sum_{g,g'} S_{gg'} p_g p_{g'}. \tag{16}$$

Then the TRI is the welfare-equivalent uniform tariff which satisfies

$$EV(\tau) = EV^{uni}(\hat{\tau}). \tag{17}$$

To make further progress, I make following simplifications:

1. Ignore cross-price terms: $S_{gg'} \approx 0$ for $g \neq g'$.
2. Locally approximate Hicksian demand by an isoelastic function, which implies

   $$S_{gg} p_g^2 \approx -\theta_g h_g p_g,$$

   where θ_g is the compensated elasticity.
3. Match only the second-order terms in the uniform-tariff and vector-tariff expressions. The idea here is that the second-order terms represent the "true" efficiency costs of the tariffs. In contrast, the first-order terms (which I'm ignoring) represent transfers from consumers to the government (who might then redistribute them back), see, e.g., Feenstra (1995).

Under these assumptions:

$$\frac{1}{2} \sum_g S_{gg} p_g^2 \hat{\tau}_g{}^2 = \frac{1}{2} \sum_g S_{gg} p_g^2 \tau_g^2. \tag{18}$$

Substituting $S_{gg} p_g^2 \approx -\theta_g h_g p_g$ yields

$$\hat{\tau}^2 \sum_g \theta_g h_g p_g = \sum_g \theta_g h_g p_g \, \tau_g^2. \tag{19}$$

And then define

$$A = \sum_g \theta_g h_g p_g, \qquad \omega_g = \frac{\theta_g h_g p_g}{A}. \tag{20}$$

Then the welfare-equivalent tariff satisfies

$$\hat{\tau} = \sqrt{\sum_g \omega_g \tau_g^2}. \tag{21}$$

This is identical in form to equation (5) in the main text. Then if compensated elasticities are identical across goods, $\theta_g = \bar{\theta}$ and invoking arguments that $h_g p_g$ is approximated by import expenditures, one has

$$\omega_g = \frac{\bar{\theta} h_g p_g}{\bar{\theta} \sum_{g'} h_{g'} p_{g'}} = \frac{M_g}{\sum_{g'} M_g} = s_g, \tag{22}$$

where s_g is the share of good g in total import expenditure. In this case, the expenditure-function approach yields the simple import-share-weighted RMS tariff:

$$\hat{\tau} = \sqrt{\sum_g s_g \tau_g^2}, \tag{23}$$

which matches equation (6) in the main text derived using the Harberger welfare-loss formulation.